Benjamin Brown

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Education

Harvey Mudd College, Claremont, CA Expected May 2028

B.S., Engineering, Emphasis in Environmental Analysis
GPA: 3.491, Dean's List
Relevant coursework: Experimental Engineering, Sustainability and Resiliency in the Built Environment, Applied Math for Engineering, Engineering Systems (with practicum), Intro to Engineering Design and Manufacturing (with lab), Chemical and Thermal Processes, Probability and Statistics for Engineers, Linear Algebra, Mechanics and Wave Motion, Intro to Computer Science (with lab), Single & Multivariable Calculus

Punahou School, Honolulu, HI June 2024

High School Diploma
GPA: 3.856, Honors
President's Award Recipient

Skills

Software & Programming: Python, MATLAB, Arduino (C++), LabVIEW, AutoCAD, SolidWorks (CAD), QGIS (Spatial Data/GIS), MS Office (Excel Financial Modeling/NPV), Adobe Creative Cloud.
Hardware & Embedded Systems: Teensy Microcontrollers, Sensor Interfacing (Thermistors, Conductivity Probes, Sonar Transducers), Analog Circuit Design, Operational Amplifiers (Op-Amps), 555 Timers, Signal Filtering, Data Logging.
Engineering Tools & Models: EPA National Stormwater Calculator, NREL PVWatts, Low-Impact Development (LID) Modeling, Techno-Economic Analysis (TEA), Net Present Value (NPV) & Payback Analysis.
Machining & Equipment: Manual Lathe, Vertical Mill, Vertical Bandsaw, Laser Cutter, 3D Printer, Heat Treatment.
Design & Quality Control: Precision Machining, Geometric Dimensioning and Tolerancing (GD&T), Dimensional Control, Engineering Drawing Interpretation, Iterative Prototyping.

Engineering Design projects

Autonomous Underwater Vehicle, *Experimental Engineering, Harvey Mudd College*
January 2026 - May 2026

- Led a 4-person engineering team to design, construct, and field-test an autonomous underwater vehicle (AUV) to measure temperature, salinity and depth data at Dana Point Harbor. Developed analog circuits utilizing operational amplifiers, 555 timers, filtering, and sensor-interface circuits using a Teensy microcontroller to capture analog signals from a thermistor, AC-driven conductivity probe, and sonar transducer. Programmed the robot's controls and onboard SD datalogging loops in Arduino C++, and developed MATLAB programs to process raw data into calibrated physical metrics.

Sustainable Wellness Resort and Eco Spa, *Sustainability and Resiliency in the Built Environment, Harvey Mudd College September 2025 – December 2025*

- Collaborated in a 3-person consulting team to engineer environmental adaptations and financial strategies for a 40-acre upscale coastal resort in a high-risk flood zone. Evaluated site hydrology using the EPA's National Stormwater Calculator to model low-impact development (LID) features, cutting baseline runoff by 50%; built Excel models to compare transit and energy strategies, executing economic trade-off analyses to determine a net present value (NPV) and payback period for proposed scenarios. Formulated multi-hazard resilience frameworks detailing elevated infrastructure, continuous load-path hurricane strapping, and HEPA-grade HVAC filtration systems.

Compost Irrigation System, *Intro to Engineering Design and Manufacturing, Harvey Mudd College*
September 2025 - December 2025

- Collaborated in a 5-person engineering team to design, prototype, and test an irrigation system for Sustainable Claremont's 4-ft compost piles, reducing watering time by ~2 hours per use and improving water efficiency. Developed SolidWorks CAD models, fabricated a PVC structural frame, and designed a 3D-printed fluid distribution mechanism; led client meetings, design reviews, and iterative prototyping

Hammer Manufacturing Project, *Intro to Engineering Design and Manufacturing, Harvey Mudd College*
September 2025 – November 2025

- Manufactured a precision steel hammer over 30+ hours to up to ±0.005in tolerance using a manual lathe, vertical mill, bandsaw, sander, and heat treatment processes. Interpreted engineering drawings, applied GD&T and dimensional control, and executed the end-to-end machining workflow from raw stock

Home PV System Project, *Personal Project, Kailua, HI*
June, 2025 – July 2025

- Used QGIS to extract rooftop geometry and spatial data, developed AutoCAD structural models, and evaluated solar PV system performance using NREL PVWatts. Performed techno-economic and environmental analysis, estimating a 6-year payback period and an annual CO_2 emissions reduction of 2,071 kg, supporting data-driven renewable energy feasibility assessment

Professional Experience

Shift Leader, Teddy's Bigger Burgers, Kailua, HI July, 2022 – August 2025

- Supervised 5 employees per shift, overseeing training of new hires, task delegation, and workflow coordination. Managed order intake and assembly, cash register operations, and end-of-shift cash, while ensuring store closing procedures, cleanliness standards, and conflict resolution

Research Experience

Summer Research, ʻIolani School, Honolulu, HI June, 2022 – August, 2022

- Collaborated with a partner to collect and analyze water samples from 8 surf locations, identifying waterborne bacteria using PCR amplification and gel electrophoresis.

Teaching & Mentorship Experience

Teacher's Assistant, Physics, Punahou School, Honolulu, HI August, 2023 – December, 2023

- Mentored and supported a cohort of 20 students, strengthening problem-solving and analytical skills through guided instruction and feedback. Facilitated lab setup, operation, and cleanup, ensuring a safe, organized, and efficient learning environment while supporting hands-on experimentation

Volunteer Camp Counselor, Punahou School, Honolulu, HI October, 2021 – October, 2022

- Led and facilitated group activities for 48 sixth-grade campers during a 3-day overnight camp, maintaining supervision, safety, and engagement. Directly responsible for a cohort of 8 campers, teaching outdoor survival skills, enforcing risk management protocols, and supporting a safe, structured camp environment